LAB 5 Part 1 : Binary Tree

[CO2]

Instructions for students:

- Complete the following methods.
- You may use Java / Python to complete the tasks.
- DO NOT CREATE a separate folder for each task just follow the given template.
- If you are using **JAVA**, then follow the **Java Template**.
- If you are using **PYTHON**, then follow the **Python Template**.

NOTE:

- **YOU CANNOT USE ANY OTHER DATA STRUCTURE OTHER THAN ARRAY UNLESS MENTIONED IN THE QUESTION.**
- **YOUR CODE SHOULD WORK FOR ANY VALID INPUTS.**

For Python: List, Negative indexing and append()
and
For Java: Static or Instance variables
are STRICTLY prohibited

Lab Tasks: 1-5
Assignment Tasks: 6-8 (3 tasks)
Total Assignment Mark: 3*5=15

Lab Task 0: Basics of Binary Tree

This is an intro task & there isn't any driver code for this

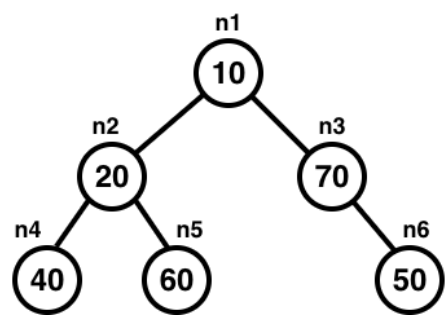
- For java, create a **separate folder** for this task and follow the instructions.
- For Python, create a **separate colab/ipynb/py** file and follow the instructions.

➔Design a TreeNode class.

- Declare three instance variables, one called **elem** (Object data type for java), one called **left** (TreeNode datatype) and another called **right** (TreeNode datatype).
- Write a constructor that has only one parameter (Object type for Java) and assigns the parameter value to the **elem** instance variable.

➔Design a Tester class (for java) / Design another code cell (for python)

- Create 6 different objects of TreeNode class. Assign values as shown in the illustration.
- Variable names should be: **n1, n2, n3, n4, n5, n6**.
- Connect the 6 TreeNodes as shown in the illustration which will form a binary tree. Here **n1** will be the root of the tree.



Now, execute the lines given below and try to understand the output. You may need pen & paper. If there are errors, try to figure out why that error occurred and how to fix it.

Note: Java & Python output might not always be the same.

JAVA	PYTHON
`System.out.println( n1.left );`	`print( n1.left )`
`System.out.println( n3.right.elem );`	`print( n3.right.elem )`
`TreeNode x = n2.left;` `System.out.println( n1.elem + x.elem );`	`x = n2.left` `print( x.elem + n6.elem )`

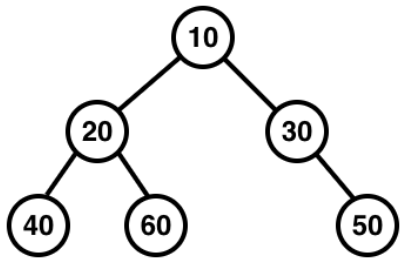
`x = new TreeNode(80);`
`n3.left = x;`
`System.out.println(n1.right.left.elem);`
	`x = TreeNode(80)`
	`n3.left = x`
	`print(n1.right.left.elem)`

`System.out.println(n1.left.right + n5.left);`	`print(n1.left.right + n5.left)`
`n1.left.right = null;` `System.out.println(n1.left.right.elem)`	`n1.left.right = None;` `print(n1.left.right.elem)`

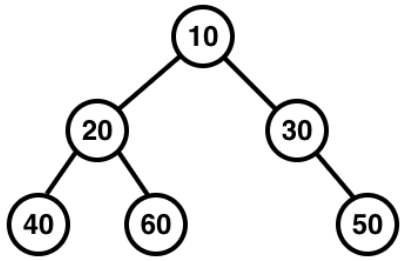
LAB TASKS

1. Basic Traversal:

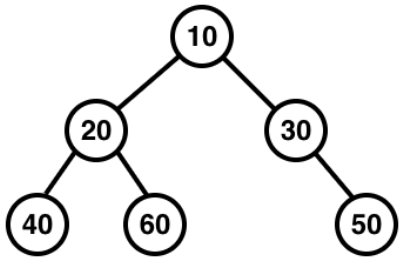
1.1. Given the **root** of a binary tree, print the tree **pre-order**.

Sample Given Tree	Sample Output
	10 20 40 60 30 50

1.2. Given the **root** of a binary tree, print the tree **post-order** and the levels of each node. The level of the root is 0. You can use a helper.

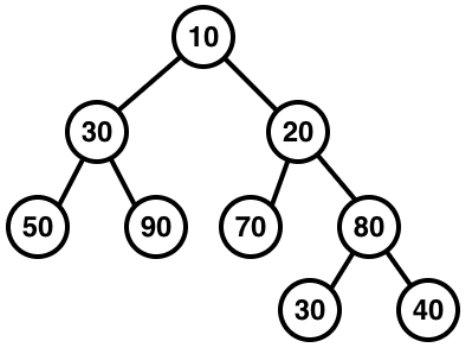
Sample Given Tree	Sample Output
	40:2 60:2 20:1 50:2 30:1 10:0

1.3. Given the **root** of a binary tree, print the tree **in-order** and the levels of each node only **if the levels are even**. The level of the root is 0.

Sample Given Tree	Sample Output
	40:2 60:2 10:0 50:2

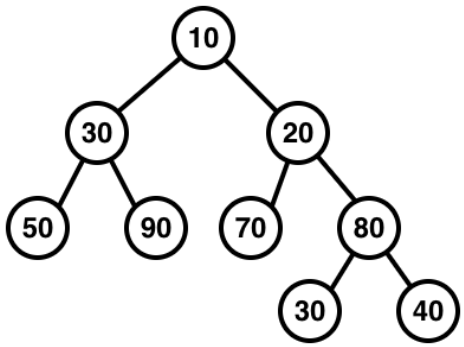
The 3 tasks above should give you a general idea about how traversing a tree works. Try to understand why the root was printed at the very beginning in 1.1 and at the very end in 1.2. You can simulate the recursive calls using pen and paper.

2. Given the **root** of a binary tree, **return** the **total number of nodes** in that tree.

Sample Given Tree	Sample Returned Value
	9

3. Summation of the tree:

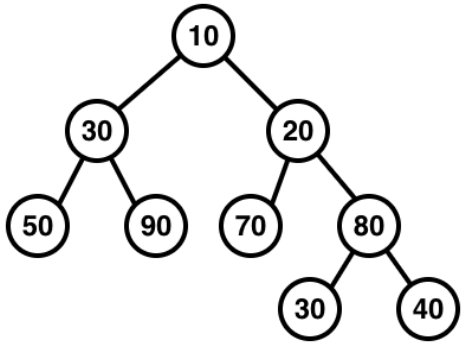
3.1. Given the **root** of a binary tree, **return** the summation of nodes in that tree. (You **won't** need helper function)

Sample Given Tree	Sample Output
	420

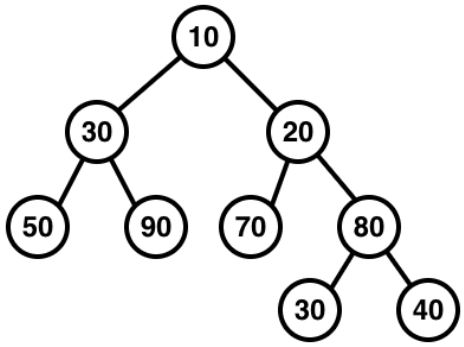
3.2. Given the **root** of a binary tree, **print** the summation of nodes in that tree. (think how to re-use 3.1 and try to understand the difference)

Sample Given Tree	Sample Returned Value
Same as above sample	Same as above sample

4. Given the **root** of a binary tree, print all the **leaf** nodes of the tree.

Sample Given Tree	Sample Output
	50 90 70 30 40

5. Given the **root** of a binary tree, return the max of all nodes.

Sample Given Tree	Sample Returned Value
	90

ASSIGNMENT TASKS

6. Subtraction of Nodes:

Write a **recursive** function **subtract_summation**() that takes the root of a binary tree as a parameter. The function will **subtract** the **summation** of the **right subtree** of the given root **from** the **summation** of the **left subtree** of the given root. Consider, the **Node** class for Binary Tree already defined with elem, left and right variables. You can use helper functions.

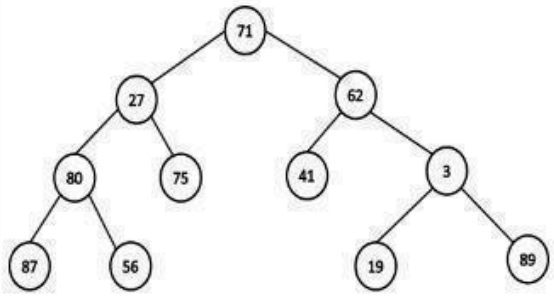
YOU CANNOT USE LIST OR DICTIONARY. You cannot use any built-in function.

Python Notation:

```
def subtract_summation(root):
    // To do
    return None
```

Function Call :

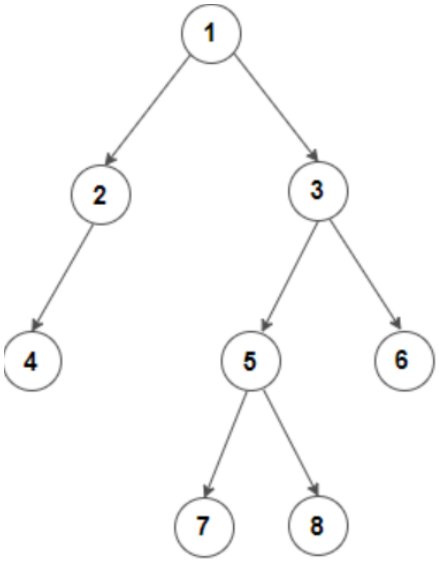
print(**subtract_summation**(root)). Here root refers to the tree below.

Sample Input	Sample Output	Explanation
	111	Summation of left subtree - summation of right subtree = (27+75+80+87+56) - (62+41+3+19+89) = 111

more tasks on the next page>>

7. Difference of Level Sum**:**

Given a Binary Tree, Write a function that finds the difference between sum of all nodes present at odd and even levels in a binary tree, i.e. sum of all odd level nodes - sum of all even level nodes.

Sample Input:	**Sample Output**	**Explanation**
	4	-1+2+3-4-5-6+7+8 = 4

more task on the next page>>

8. Swap Children Nodes:

Write a **recursive** function **swap_child()** that takes the root of a binary tree, node's level and a number M as a parameter. The function will swap the left and right children of all the nodes at level M and above. Here, 0 < M < height of the tree (root's height). Consider, the Node class for Binary Tree already defined with elem, left and right variables. **YOU CANNOT USE LIST OR DICTIONARY, any built-in function, global variables.**

Python Notation:

```
def swap_child(root, level, M):
        # To do
```

Function Call :

swap_child(root, 0, 2). Here root refers to the tree below.

Input Tree	Resulting Tree	Explanation
A / \ B C / \ \ D E F / \ / / G H I J	A / \ C B / / \ F E D / / / \ J I G H	Here M = 2 and all the nodes from level 2 and above are swapped left with right. Here above means the level that situated at a higher position of the tree